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BEVERAGE DISPENSING APPARATUS

Abstract

A beverage dispensing apparatus includes: an extraction unit extracting a beverage as an ingredient and hot water are fed; a nozzle that supplies the beverage extracted by the extraction unit to a container; and an addition-hot-water supply unit that supplies the hot water to the nozzle as addition hot water forming the beverage, without passing through the extraction unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of International Application No. PCT/JP2023/032712, filed on Sep. 7, 2023 which claims the benefit of priority of the prior Japanese Patent Application No. 2023-015256, filed on Feb. 3, 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] The present invention relates to a beverage dispensing apparatus.

[0003] In the related art, in beverage dispensing apparatuses that supply beverages to containers, an extraction unit arranged inside an apparatus main body extracts a beverage from a ground ingredient and hot water, and the extracted beverage is then dispensed into a container such as a cup (refer to Japanese Laid-open Patent Publication No. 2007-094869).

[0004] In the beverage dispensing apparatus described above, a conduit for supplying hot water for dilution (hereinafter, also referred to as “dilution hot water”) to a container is provided. This conduit supplies hot water stored in a hot water tank as dilution hot water to a dedicated dilution water nozzle without passing through an extraction unit, and the dilution hot water is then dispensed into the container from the nozzle.

SUMMARY

[0005] There is a need for providing a beverage dispensing apparatus that enables preferable cleaning of a nozzle that dispenses a beverage extracted by an extraction unit.

[0006] According to an embodiment, a beverage dispensing apparatus according to the present invention includes an extraction unit extracting a beverage as an ingredient and hot water are fed, and that supplies the beverage extracted by the extraction unit to a container through a nozzle. Further, the beverage dispensing apparatus further includes: an addition-hot-water supply unit that supplies the hot water to the nozzle as addition hot water forming the beverage, without passing through the extraction unit.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1 is a perspective view illustrating an external configuration of a beverage dispensing apparatus that is an embodiment of the present invention;

[0008] FIG. 2 is a exploded perspective view of the beverage dispensing apparatus that is an embodiment of the present invention;

[0009] FIG. 3 is a schematic diagram schematically illustrating respective components of the beverage dispensing apparatus that is an embodiment of the present invention;

[0010] FIG. 4 is a schematic diagram schematically illustrating a connection state of a beverage dispensing conduit and an addition-water supply conduit in FIG. 3;

[0011] FIG. 5 is an explanatory diagram illustrating an extraction procedure of a beverage by the beverage dispensing apparatus illustrated in FIG. 1 to FIG. 3;

[0012] FIG. 6 is an explanatory diagram illustrating an extraction procedure by the beverage dispensing apparatus illustrated in FIG. 1 to FIG. 3;

[0013] FIG. 7 is an explanatory diagram illustrating an extraction procedure by the beverage dispensing apparatus illustrated in FIG. 1 to FIG. 3;

[0014] FIG. 8 is an explanatory diagram illustrating an extraction procedure by the beverage dispensing apparatus illustrated in FIG. 1 to FIG. 3;

[0015] FIG. 9 is an explanatory diagram illustrating an extraction procedure by the beverage dispensing apparatus illustrated in FIG. 1 to FIG. 3;

[0016] FIG. **10** is an explanatory diagram illustrating an extraction procedure by the beverage dispensing apparatus illustrated in FIG. **1** to FIG. **3**; and

[0017] FIG. **11** is an explanatory diagram illustrating an extraction procedure by the beverage dispensing apparatus illustrated in FIG. **1** to FIG. **3**.

DETAILED DESCRIPTION

[0018] In the related-art beverage dispensing apparatus, because the dilution hot water has been supplied from a dedicated dilution hot water nozzle that is separate from a nozzle dispensing a beverage extracted by the extraction unit, it has been impossible to clean the nozzle.

[0019] Hereinafter, exemplary embodiments of a beverage dispensing apparatus according to the present invention will be explained in detail with reference to the accompanying drawings.

[0020] FIG. **1** to FIG. **3** respectively illustrate the beverage dispensing apparatus that is an embodiment of the present invention. FIG. **1** is a perspective view illustrating an external configuration, FIG. **2** is an exploded perspective view, and FIG. **3** is a schematic diagram schematically illustrating respective components.

[0021] The beverage dispensing apparatus illustrated herein is, for example, a coffee machine installed in stores such as convenience stores, that performs extraction processing such as grinding coffee beans and dripping, and supplies beverages such as coffee to a container C, such as a cup. Such a beverage dispensing apparatus includes an apparatus main unit **1**.

[0022] The apparatus main unit **1** is configured to include a main unit cabinet **10** and a front door **20**. The main unit cabinet **10** has a substantially rectangular prism shape with an opening (hereinafter, also referred to as “front opening”) on its front not illustrated. Inside this main unit cabinet **10**, a beverage production unit **10A** that makes a beverage (for example, coffee) and a control unit **10B** are arranged.

[0023] The front door **20** is a door unit that is in a size large enough to close the front opening of the main unit cabinet **10**. This front door **20** is arranged swingably around a center axis of an axial portion not illustrated that extends along a vertical direction at a left edge portion on a front side of the main unit cabinet **10**, and is capable of opening and closing the front opening of the main unit cabinet **10**.

[0024] In the front door **20**, its front side forms a customer-facing side, and is provided with a display unit **21**, a beverage dispensing unit **22**, and an opening/closing door **23**.

[0025] The display unit **21** is constituted of, for example, a liquid crystal touch panel, and is configured to display various kinds of information according to an instruction provided by the control unit **10B**, and to enable input operation, such as touch operation. The display unit **21** transmits a purchase signal to the control unit **10B** when an input operation, such as a touch operation, is performed.

[0026] The beverage dispensing unit **22** is arranged below the display unit **21**, and includes a nozzle **22a** and a stage **22b**. The nozzle **22a** discharges a beverage produced by the beverage production unit **10A** downward. More specifically, the nozzle **22a** is a diffuser nozzle that stores the beverage sent to itself once, and then discharges it downward. The stage **22b** is arranged in an area below the nozzle **22a**. This stage **22b** is to put the container C thereon.

[0027] The opening/closing door **23** is made of a translucent material such as transparent resin, and is in a size large enough to close an inlet **22c** of the beverage dispensing unit **22**. The opening/closing door **23** is pivotally supported by the front door **20** at its left end portion, and is swingable along the front-rear direction. That is, the opening/closing door **23** is swingable along the front-rear direction in such a manner that it moves toward and away from the beverage dispensing unit **22**. The opening/closing door **23** can close the inlet **22c** of the beverage dispensing unit **22** when it swings rearward to approach the beverage dispensing unit **22**, and can open the inlet **22c** of the beverage dispensing unit **22** when it swings forward to move away from the beverage dispensing unit **22**.

[0028] The beverage production unit **10A** described above will be explained. The beverage

production unit **10A** includes an ingredient storage unit **31**, a grinder **33**, a hot-water supply unit **35**, an extraction unit **37**, a collection bucket **39**, a compressed-air supply unit **41**, a drainage unit **43**, and an addition-hot-water supply unit **45**.

[0029] The ingredient storage unit **31** is to store roasted coffee beans, which are a beverage ingredient, and is arranged in such a manner that a part thereof extends upward from a top panel of the main unit cabinet **10**. This ingredient storage unit **31** includes an ingredient-supply driving unit **311**. The ingredient-supply driving unit **311** operates when a drive instruction is given from the control unit **10B**, and dispenses a predetermined amount of coffee beans specified in the drive instruction.

[0030] The grinder **33** is a so-called mill, and operates when a drive instruction is given by the control unit **10B**. This grinder **33** is arranged at an area below the ingredient storage unit **31**, and is connected to the ingredient storage unit **31** through an ingredient chute **32**.

[0031] The grinder **33** grinds, when operating, coffee beans that has been dispensed from the ingredient storage unit **31** and guided thereto by the ingredient chute **32**, and feeds the ground coffee beans (hereinafter, also referred to as ground coffee) into the extraction unit **37** through a powder chute **34**.

[0032] The hot-water supply unit **35** supplies hot water to the extraction unit **37**, and is constituted of a hot water tank **351**, a metering pump **352**, a sub-tank **353**, a hot-water supply pump **354**, and a check valve **355** sequentially connected to a hot-water supply conduit **356** constituted by hot-water supply piping.

[0033] The hot water tank **351** heats water, such as tap water, supplied from a water supply means not illustrated using a heater **351a**, and stores it as hot water. The metering pump **352** is constituted by a non-positive displacement pump, such as a centrifugal pump. The metering pump **352** is driven in response to an instruction from the control unit **10B**, and when driven, sends a specified amount of hot water from the hot water tank **351** to the sub-tank **353**. The sub-tank **353** is smaller in volume than the hot water tank **351**, and temporarily stores the hot water sent by the metering pump **352**. The hot-water supply pump **354** is constituted by a positive displacement pump, such as a gear pump. This hot-water supply pump **354** is driven in response to an instruction from the control unit **10B**, and when driven, compresses the hot water in the sub-tank **353** and sends it to the extraction unit **37**. The hot-water supply pump **354** is configured such that its discharge amount is larger than the discharge amount of the metering pump **352**. The check valve **355** is a valve body that allows hot water sent from the hot-water supply pump **354** to pass toward the extraction unit **37**, while restricting passage of hot water from the extraction unit **37** to the sub-tank **353**. Although not explicitly illustrated in the drawing, the check valve **355** is arranged in a state thermally connected to the hot water tank **351**.

[0034] The extraction unit **37** is a so-called brew unit, and extracts coffee from the ground coffee fed from the grinder **33** through the powder chute **34** and hot water supplied by the hot water supply unit **35**.

[0035] To the extraction unit **37** as described, a beverage supply conduit **38** is connected. The beverage supply conduit **38** is constituted of a single beverage supply conduit, or constituted by connecting multiple beverage supply conduits, and supplies coffee extracted by the extraction unit **37** to the nozzle **22a**.

[0036] In this beverage supply conduit **38**, a first pinch valve **381** is arranged in the middle. This first pinch valve **381** opens and closes in response to an instruction given by the control unit **10B**, and allows fluid to flow when it is in an open state, while restricting passage of fluid when it is in a closed state.

[0037] The extraction unit **37** described above includes a cylinder **371**, a lid **372**, and a filter block **373**. The cylinder **371** has a substantially cylindrical shape, and is arranged detachably with respect to the main unit cabinet **10**.

[0038] The lid **372** moves in such a manner that it approaches and retracts from an upper opening

of the cylinder **371** as the lid **372**, such as a motor, drives in response to an instruction given by the control unit **10B**, thereby opening and closing the upper opening of the cylinder **371**. Although not illustrated, in the lid **372**, a hole that allows ground coffee supplied through the power chute **34** to be put into the cylinder **371**, and a hole that allows hot water supplied by the hot-water supply unit **35** to be put into the cylinder are formed.

[0039] The filter block **373** is arranged in an area below the cylinder **371**, and is connected to the beverage supply conduit **38** described above. This filter block **373** moves in a vertical direction in such a manner that it approaches and retracts from the cylinder **371** as a block drive unit **373a**, such as a motor, drives in response to an instruction given by the control unit **10B**.

[0040] The extraction unit **37** includes a paper roller **50**. The paper roller **50** holds a paper filter **56** that is drawn out from a filter roll **55** that is rotatably supported and housed in a filter storage unit **47**. The paper roller **50** is constituted of a drive roller **51** and a driven roller **52**.

[0041] The drive roller **51** rotates around its center axis when a roller driving unit **51a**, which is a driving source, drives in response to an instruction from the control unit **10B**.

[0042] The driven roller **52** rotates around its center axis as the drive roller **51** rotates, by sandwiching the paper filter **56** between a portion of an outer periphery thereof and a corresponding portion of an outer periphery of the drive roller **51**.

[0043] The collection bucket **39** is arranged in an area below the extraction unit **37**, and stores an extraction residue K (refer to FIG. **11**) generated during extraction of a beverage in the extraction unit **37**, together with the paper filter **56**.

[0044] The compressed-air supply unit **41** includes a compressed-air supply conduit **411**, an air pump **412**, a second pinch valve **413**, and a third pinch valve **414**.

[0045] The compressed-air supply conduit **411** is formed by connecting multiple compressed-air supply conduits, and its one end is connected to the lid **372** and the other end is connected to a middle portion of the beverage supply conduit **38**.

[0046] The air pump **412** is provided in the compressed-air supply conduit **411**. This air pump **412** is driven in response to an instruction from the control unit **10B** and compresses air to send it out.

[0047] The second pinch valve **413** is arranged on one end side (the lid **372** side) relative to the air pump **412** in the compressed-air supply conduit **411**. This second pinch valve **413** opens and closes in response to an instruction given by the control unit **10B**, and allows passage of fluid when it is in an open state, while restricting passage of fluid when it is in the closed state.

[0048] The third pinch valve **414** is arranged on the other side to the air pump **412** (on the beverage supply conduit **38** side) in the compressed-air supply conduit **411**. This third pinch valve **414** opens and closes in response to an instruction given by the control unit **10B**, and allows passage of fluid when it is in an open state, while restricting passage of fluid when it is in the closed state.

[0049] The drainage unit **43** includes a drainage conduit **431** and a fourth pinch valve **432**. The drainage conduit **431** is formed by connecting one or more drainage pipes, and is arranged in such a manner that it branches from a middle portion of the beverage supply conduit **38** and extends into an area above the collection bucket **39**.

[0050] The fourth pinch valve **432** is arranged in the drainage conduit **431**. This fourth pinch valve **432** opens and closes in response to an instruction given by the control unit **10B**, and allows passage of fluid such as waste liquid when it is in an open state, while it restricts passage of fluid such as waste liquid when it is in a closed state. Thus, the collection bucket **39** stores the extraction residue K generated during extraction of a beverage in the extraction unit **37**, the paper filter **56** used in the extraction of the beverage, and waste liquid generated by the cleaning of the extraction unit **37**.

[0051] The addition-hot-water supply unit **45** includes an addition-hot-water supply conduit **451** and an addition-hot-water pump **452**. The addition-hot-water supply conduit **451** is formed by connecting multiple addition-hot-water supply conduits, and its one end is connected to a portion between the sub-tank **353** and the hot-water supply pump **354** in the hot-water supply conduit **356**,

and the other end is connected to a junction point P between the first pinch valve **381** and the nozzle **22a** in the beverage supply conduit **38**.

[0052] Connection of the addition-hot-water supply conduit **451** and the beverage supply conduit **38** will be explained. As illustrated in FIG. 4, between the first pinch valve **381** and the nozzle **22a** in the beverage supply conduit **38**, an inclined extending portion **38a** that gradually extends upward toward the nozzle **22a** is provided, and the junction point P described above is formed in this inclined extending portion **38a**.

[0053] On the other hand, the addition-hot-water supply conduit **451** inclines gradually downward as it approaches the inclined extending portion **38a**, and is connected to the junction point P in such a manner that an angle θ formed with the inclined extending portion **38a** is an acute angle, that is, an acute angle with respect to a direction in which coffee (beverage) is supplied in the inclined extending portion **38a**.

[0054] The addition-hot-water pump **452** is arranged on the addition-hot-water supply conduit **451**. This addition-hot-water pump **452** is constituted by a positive displacement pump, such as a diaphragm pump, and is driven in response to an instruction given by the control unit **10B**. The addition-hot-water pump **452** sends out hot water in the sub-tank **353** as addition hot water forming coffee to the nozzle **22a** through the addition-hot-water supply conduit **451** when driving, while it restricts sending addition water to the nozzle **22a** through the addition-hot-water supply conduit **451** when the driving is stopped.

[0055] The control unit **10B** comprehensively controls operation of the respective components of the beverage dispensing apparatus according to a program or data stored in the storage unit not illustrated.

[0056] The control unit **10B** may be implemented by causing a processing device, such as a central processing unit (CPU), to execute a program, that is, by software, may be implemented by hardware, such as an integrated circuit (IC), or may be implemented by using software and hardware.

[0057] In the beverage dispensing apparatus configured as described above, as illustrated in FIG. 5 to FIG. 11, coffee is supplied to the container C placed on stage **22b** of the beverage dispensing unit **22**. As a premise, it is assumed that hot water at a predetermined temperature is generated and stored in the hot water tank **351**, and the first pinch valve **381**, the second pinch valve **413**, and the fourth pinch valve **432** are in the closed state, while the third pinch valve **414** is in the open state.

[0058] As the display unit **21** is operated by touch operation by a user, the control unit **10B** to which a purchase signal of a selected beverage is given moves the filter block **373** upward as illustrated in FIG. 5, and then sends a drive instruction to the ingredient-supply driving unit **311** to feed an amount of coffee beans suitable for the beverage to the grinder **33**, and sends a drive instruction to the grinder **33** to grind the coffee beans and put into the extraction unit **37**.

Thereafter, it stops driving of the ingredient-supply driving unit **311**.

[0059] The control unit **10B** drives the metering pump **352** to send a specific amount of hot water from the hot water tank **351** to the sub-tank **353**, while driving the hot-water supply pump **354** to compress the hot water in the sub-tank **353** and send it to the extraction unit **37**, thereby putting hot water to the extraction unit **37**. Thereafter, the driving of the metering pump **352** and the hot-water supply pump **354** is stopped.

[0060] The control unit **10B** then drives the air pump **412** to supply compressed air (compressed air for stirring) to the cylinder **371** to stir (forcibly stir) the ground coffee and hot water using a part of the beverage supply conduit **38** as illustrated in FIG. 6. Thereafter, the driving of the air pump **412** is stopped.

[0061] The control unit **10B** that has forcibly stirred the ground coffee and hot water as described moves the lid **372** to close the upper opening of the cylinder **371**, brings the third pinch valve **414** into the closed state, brings the first pinch valve **381** and the second pinch valve **413** into the open state, and drives the air pump **412**. The fourth pinch valve **432** is maintained in the closed state.

[0062] Thus, as illustrated in FIG. 7, coffee is extracted by supplying compressed air (compressed air for extraction) to the cylinder **371**, and the extracted coffee is supplied to the nozzle **22a** through the beverage supply conduit **38**, and is dispensed to the container C from the nozzle **22a**.

[0063] The control unit **10B** then drives the addition-hot-water pump **452**. Thus, as illustrated in FIG. 8, hot water in the sub-tank **353** is sent out to the nozzle **22a** through the addition-hot-water supply conduit **451** as the addition hot water, and the addition hot water is dispensed to the container C as an element forming coffee.

[0064] After the discharge of the coffee extracted by the extraction unit **37** into the container C is completed, the control unit **10B** stops the driving of the air pump **412** but continues the driving of the addition-hot-water pump **452**. Thus, as illustrated in FIG. 9, the hot water in the sub-tank **353** is discharged as addition hot water from the nozzle **22a** into the container C through the addition-hot-water supply conduit **451**.

[0065] When a predetermined amount of coffee is discharged and supplied into the container C, the control unit **10B** stops the driving of the addition-hot-water pump **452**. Thus, the user can take out the container C from the beverage dispensing unit **22** by swinging the opening/closing door **23** in the opening direction.

[0066] Thereafter, the control unit **10B** moves the lid **372** to open the upper opening of the cylinder **371**, to release the internal pressure in the cylinder **371**. The control unit **10B** then supplies hot water from the hot-water supply unit **35** to the extraction unit **37**, to clean an interior of the cylinder **371**, and brings the fourth pinch valve **432** into the open state, to discharge remaining water in the extraction unit **37** as waste fluid into the collection bucket **39** through the drainage conduit **431** as illustrated in FIG. 10.

[0067] The control unit **10B** that has discharged the waste fluid as described brings the fourth pinch valve **432** into the closed state as illustrated in FIG. 11, and moves the filter block **373** downward. The control unit **10B** makes the paper roller **50** to feed out the paper filter **56** by a predetermined amount, causes the collection bucket **39** to collect the residue K generated by extraction of coffee together with the paper filter **56**, and ends dispensing of coffee this time.

[0068] As explained above, according to the beverage dispensing apparatus that is an embodiment of the present invention, hot water is supplied to the nozzle **22a** as addition hot water by the addition-hot-water supply unit **45** without passing through the extraction unit **37**. Therefore, the nozzle **22a** that dispenses a beverage extracted by the extraction unit **37** can be cleaned with the addition hot water supplied as a beverage. Accordingly, the nozzle **22a** that dispenses a beverage extracted by the extraction unit **37** can be preferably cleaned.

[0069] Particularly, because the control unit **10B** drives the addition-hot-water pump **452** also after extraction of coffee by the extraction unit **37**, the nozzle **22a** can be cleaned reliably with the addition hot water.

[0070] Moreover, according to the beverage dispensing apparatus, because the addition hot water constituting a beverage is supplied to the container C, the addition water is not mixed with ground coffee, and there is no risk of extracting components of strange flavor or the like, thereby suppressing degradation of the quality of the coffee supplied to the container C.

[0071] Furthermore, according to the beverage supply apparatus, because the addition-hot-water supply conduit **451** is gradually inclined downward as it approaches the inclined extending portion **38a** of the beverage supply conduit **38** and is connected to form an acute angle with respect to the direction in which coffee is supplied in the inclined extending portion **38a**, it is possible to prevent coffee from flowing backward to the addition-hot-water supply conduit **451** when coffee is supplied to the nozzle **22a** through the beverage supply conduit **38**.

[0072] As above, an exemplary embodiment of the present invention has been explained, but the present invention is not limited thereto, and various modifications can be made.

[0073] In the embodiment described above, the addition-hot-water pump **452** is driven also after extraction of coffee, but in the present invention, the driving timing can be arbitrarily set.

[0074] Moreover, in the present invention, even when a beverage discharged from the nozzle 22a is only coffee extracted by the extraction unit 37 and supply of addition hot water from the addition-hot-water supply unit 45 is not necessary, the addition-hot-water pump 452 may be driven.

[0075] This enables to supply remaining hot water or the like in the addition-hot-water supply conduit 451 to the nozzle 22a.

[0076] According to the present invention, because an addition-water supply unit supplies hot water as addition hot water that forms a beverage to a nozzle without passing through an extraction unit, the nozzle that dispenses the beverage extracted by the extraction unit can be cleaned by the addition water supplied as the beverage. Therefore, an effect of enabling preferable cleaning of the nozzle that dispenses the beverage extracted by the extraction unit is produced.

[0077] Although the disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A beverage dispensing apparatus comprising: an extraction unit extracting a beverage as an ingredient and hot water are fed; a nozzle that supplies the beverage extracted by the extraction unit to a container; and an addition-hot-water supply unit that supplies the hot water to the nozzle as addition hot water forming the beverage, without passing through the extraction unit.
 2. The beverage dispensing apparatus according to claim 1, wherein the addition-hot-water supply unit includes an addition-hot-water supply conduit to supply the addition hot water to the nozzle; and an addition-hot-water pump that is arranged in the addition-hot-water supply conduit, and that sends the addition hot water to the nozzle through the addition-hot-water supply conduit when driven, while restricting sending the addition hot water to the nozzle through the addition-hot-water supply conduit when the driving is stopped.
 3. The beverage dispensing apparatus according to claim 2, further comprising a control unit that drives the addition-hot-water pump also after extraction of the beverage by the extraction unit.
 4. The beverage dispensing apparatus according to claim 2, further comprising a beverage supply conduit that supplies a beverage extracted by the extraction unit to the nozzle, and that has an inclined extending portion that extends in a gradually inclined manner upward toward the nozzle, wherein the addition-hot-water supply conduit is connected in such a manner that it gradually inclined downward as it approaches the inclined extending portion, and forms an acute angle with respect to a direction in which a beverage is supplied in the inclined extending portion.
 5. The beverage dispensing apparatus according to claim 3, further comprising a beverage supply conduit that supplies a beverage extracted by the extraction unit to the nozzle, and that has an inclined extending portion that extends in a gradually inclined manner upward toward the nozzle, wherein the addition-hot-water supply conduit is connected in such a manner that it gradually inclined downward as it approaches the inclined extending portion, and forms an acute angle with respect to a direction in which a beverage is supplied in the inclined extending portion.
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